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PROFILE

Senior Data Scientist & AI/ML Engineer with 6+ years of experience designing, training, and deploying production Machine Learning, Deep Learning, and Generative AI systems. Specialized in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, and multi-agent AI architectures — from training Transformer models from scratch to fine-tuning LLMs and shipping systems that scale. Delivered 50+ AI/ML projects for clients across the USA, UK, Middle East, and Europe.

PROFESSIONAL EXPERIENCE

Senior Data Scientist Jul 2025 – Present | Islamabad, Pakistan
Appedology

Lead Data Scientist Feb 2025 – Jul 2025 | Islamabad, Pakistan
AlphaSquad 🔗

- **Built and fine-tuned** a persona-driven crypto Twitter agent (Mind of Pepe) by scraping and cleaning tweets from the @aixbt agent and adapting a LLaMA model to mimic its style, achieving 98% signal accuracy and viral engagement.
- **Shipped a gated** terminal integrating AI-based signal generation, crypto sentiment analysis, and technical analysis using real-time CoinMarketCap data, delivering actionable insights to meme coin traders.
- **Integrated a Retrieval-Augmented Generation (RAG) system** with the agent and terminal, enabling contextual understanding by pulling live crypto data and user behavior patterns.
- **Built a full AI-powered receptionist system** using n8n, Telnyx, and ElevenLabs for Jumper Media, enabling intelligent call handling, event scheduling, follow-ups, and outbound calls.
- **Integrated real-time speech-to-text transcription and RAG** with webhook-driven context from customer emails and call logs, improving personalization and relevance.
- **Optimized** system performance, reducing response latency to 850ms for a seamless, human-like user experience.

Data Scientist Aug 2024 – Feb 2025 | Lahore, Pakistan
MetaViz Pro 🔗

- Delivered Computer Vision, Deep Learning, and Natural Language Processing (NLP) systems, including engineering Transformers, Large Language Models (LLMs), and AI agents from scratch; leveraged Data Science to solve business problems.
- **Trained two Transformer-based models** from scratch using GPT-NeoX and PyTorch: a 5M-parameter text generation model (74% accuracy) and a 19M-parameter Python/Java code completion model (67% accuracy).
 - **Built a real-time yoga pose estimation system** using computer vision, comparing extracted pose features against a reference database and providing corrective feedback.
 - **Engineered a facial recognition attendance system with PPE** compliance monitoring, triggering alerts and identifying non-compliant workers in real time.
 - **Optimized a Convolutional Neural Network (CNN)** for brain tumor detection, achieving 98% accuracy with a 5-layer architecture — matching EfficientNet-level performance with a far more compact model.
 - **Built an AI agent system** in LangGraph to score AI job-replacement risk, perform skills gap analysis, and recommend career pathways.
 - **Managed and cleaned complex financial data**, engineered machine learning models for insights, and analyzed trends to support strategic financial decisions.

JMM Technologies 🔗

- Data Scientist** Sep 2023 – Aug 2024 | Peshawar, Pakistan
Created Computer Vision, Deep Learning, and NLP chatbot systems using Large Language Models (LLMs); employed Data Science to solve business problems.
- **Engineered an AI web app** using GPT-4o to detect translation errors across Spanish, French, and Arabic documents against English source text, improving accuracy from 50% to 90% through advanced prompt engineering and reducing costs 40% via token management and Python regex.

- **Led a US company's end-to-end system** deploying regression algorithms for football score predictions, including cleansing data from a third-party API for optimal input quality.
- **Built an AI system for generating** custom LLM-based knowledge assistants from YouTube videos, PDFs, and website content using the OpenAI Assistant API, LangChain for video transcription, and web scraping — enabling RAG systems and code interpreters.
- **Fine-tuned YOLOv8 and YOLOv5** for an American Sign Language detection system (MVP stage) for Saudi Airport Assistance, achieving high detection accuracy while resolving FPS lag.
- **Built an advanced RAG application** using Query Expansion and Cross-Encoder Re-ranking, improving retrieval accuracy.
- **Used BeautifulSoup, Selenium, and Octoparse** for large-scale web scraping of real-time data from authorized sources.
- **Utilized time series modeling** (ARIMA, SARIMA, Facebook Prophet) to forecast temperature and inflation trends.
- **Deployed ML pipelines on AWS** (EC2, Lambda, S3, Cloud9) for scalable data storage and transformation.
- **Built sales dashboards in Power BI**, turning complex datasets into actionable insights for decision-making.
- **Delivered data cleaning and analysis** for an e-recycling company project, generating actionable business insights.

Machine Learning Engineer Associate

Nov 2022 – May 2023 | Quetta, Pakistan

Government Innovation Lab (United Nation Development Program UNDP) 

- **Led a 3-person team** (including 1 junior team member) in developing a facial-recognition-based Smart Attendance System achieving 94% accuracy, replacing the department's manual fingerprint-based attendance process.
- **Contributed to a Smart Vehicle Counting system** using machine learning algorithms for precise, automated vehicle counting.
- **Collaborated with cross-functional teams** as ICT Developer Fellow, integrating emerging technologies into government infrastructure.

Data Science Intern

Jun 2022 – Jul 2022 | Quetta, Pakistan

British Airways

- **Analyzed a dataset** of 20,000+ booking records to identify patterns behind a 27% ticket cancellation rate, informing targeted mitigation strategies.
- **Designed predictive models** using machine learning algorithms to optimize operational performance and forecast trends.
- **Translated complex data** findings into actionable insights and recommendations, contributing to improved profit margins and operational efficiency.

Research Assistant

Jan 2021 – Dec 2021 | Quetta, Pakistan

University of Balochistan (Computer Science Department)

- **Trained and evaluated** a Gender Prediction model achieving 99% accuracy under PhD supervision as part of an academic research initiative.
- **Implemented and compared** four detection models (Faster R-CNN, R-CNN, SSD, YOLOv8) for license plate recognition, achieving up to 86% accuracy and contributing to an academic publication.

SKILLS

Core Competencies

Machine Learning, Deep Learning, Large Language Models (LLMs), Generative AI, Retrieval-Augmented Generation (RAG), Computer Vision, Natural Language Processing (NLP), AI Agents, MLOps

ML / DL Frameworks & Libraries

TensorFlow, PyTorch, Keras, Scikit-Learn, Ultralytics (YOLO), OpenCV, Pillow, Pandas, NumPy, Transformers, LangChain, LangGraph, CrewAI, LlamaIndex, HuggingFace, Pydantic, NLTK, PySpark

Visualization & BI

Matplotlib, Seaborn, Plotly, Power BI

Hardware & IoT

NVIDIA Jetson Nano, Raspberry Pi, Arduino, OpenCV OAK-D

Programming Languages

Python, C++, JavaScript, MySQL, PostgreSQL

MLOps & Data Tools

MLflow, Weights & Biases, DVC, dbt, Snowflake, Git

Web Frameworks

Django, FastAPI, Flask, Streamlit, Gradio, Dash, HTML, CSS, Selenium, BeautifulSoup, Octoparse

Cloud & DevOps

AWS (EC2, Lambda, S3, Kinesis), Azure (OpenAI, AI Search, Document Intelligence, App Services, Container Apps, Function Apps), Google Cloud Platform, IBM Cloud, DigitalOcean, Docker, Kubernetes, GitHub Actions, CI/CD, Airflow

PROJECTS

Transformers From Scratch

Tech Stack: Python, GPT-NEOX, CodeXGLUE, PyTorch

- **Constructed a Transformer-based** text generation model (5M parameters) using GPT-NeoX and PyTorch, achieving 74% accuracy in generating contextually relevant text.
- **Created a Transformer-based** code completion model for Python and Java (19M parameters), achieving 67% accuracy.
- **Fine-tuned and evaluated** both models on task-specific datasets, achieving strong performance with compact architectures.

ProofLingo

Tech Stack: OpenAI, Langchain

- **Built an AI system** to detect contextual, missing, consistency, and translation errors by comparing source and target language documents using GPT-4, achieving over 90% accuracy.
- **Reduced system costs** by 40% through token management and prompt engineering.
- **Created AI tools** for text formatting, bilingual document queries, tone adaptation, and rephrasing; collaborated with the backend team to build RESTful APIs for CRUD operations.

Sign Language Detection

Tech Stack: TensorFlow, Keras, OpenCv, Ultralytics, OAK-D,

- **Fine-tuned** YOLOv8, LSTM, and CNN1D models for sign language detection as part of a Saudi government initiative.
- **Integrated the models** with ChatGPT to enhance language processing capabilities.
- **Combined computer vision and language** understanding technologies to deliver an effective sign language detection system.

Smart Attendance System

Tech Stack: Scikit-Learn, OpenCv, Numpy, Flask, HTML, CSS, Ajax

- **Built a facial recognition**-based attendance system using SVM and KNN models for accurate individual recognition.
- **Deployed the system** within the Planning and Development Department, improving operational efficiency and security.

Smart Vehicle Counting System

Tech Stack: OpenCv, LabelImg, JetsonNano

- **Collected, annotated,** and cleaned vehicle image data to build a robust training dataset.
- **Optimized a YOLOv8 model** on vehicle data, improving object detection accuracy.

Self-Driving Car

Tech Stack: Raspberry Pi, Arduino, OpenCv

- **Successfully executed a FYP focused** on a Self-Driving Vehicle, integrating cutting-edge IoT and AI technologies.
- **Demonstrated proficiency** in merging innovative technologies to create an advanced autonomous vehicle system.
- **Applied practical skills** and theoretical knowledge to contribute to the forefront of self-driving vehicle development.

License Plate Detection

Tech Stack: Python, TensorFlow, OpenCv, Keras, Numpy

- **Implemented License Plate** Detection for a research paper project under PhD supervision.
- **Trained four models** (Faster RCNN, RCNN, SSD, YOLOv8) to achieve comprehensive license plate recognition.
- **Contributed to advancing** the field through hands-on application of diverse detection models in a real-world research setting.

EDUCATION

Bachelor's in Computer Science

University Of Balochistan

CGPA 3.36/4.00

Aug 2018 – Dec 2022 | Quetta, Pakistan

AWARDS

IEEE Workshop

IEEE

Training program in university from IEEE. In Award I get Certificate and also IEEE publication account for free.